

UNIVERSIDAD DE GUADALAJARA

CENTRO UNIVERSITARIO DE LOS VALLES



Maestría en Ingeniería de Software

Course: Administración de la configuración del software

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Change Request CR-01 v2

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CHANGE REQUEST CR-01 v2

StudyGo — Change analysis and impact

1. CLIENT REQUEST

The school asks for a new statistics module about all the courses by group of the whole school, not only the one of a teacher with their group.

The module must show the statistics of each course and let the school know if the students are learning or not. The school also wants to know if the use of the app has an impact on the learning of the students.

2. INTERPRETATION OF THE REQUEST

The request was received in general terms. These assumptions were made to analyze it:

1. The client needs to see the progress of the students by course and by group of the whole school.
2. The client needs to know if the use of the app improves learning.

These assumptions would be confirmed with the client before signing the change.

3. CONFIGURATION IDENTIFICATION

3.1 New requirements

The change creates six new functional requirements and modifies one existing requirement.

Table 1. New requirements.

ID	NAME	DESCRIPTION
RF-34	Academic office role	The system has a role that can see the information of the whole school
RF-35	Statistics by course	The academic office sees the average of correct answers of each course
RF-36	Comparison between groups	The academic office compares the groups of the same course
RF-37	Save the date of the result	The system saves the date of each homework that is answered
RF-38	Learning over time	The system shows how the average of correct answers changes over time
RF-39	Students with low results	The academic office sees the list of students with a low average in one or more subjects

Table 2. Modified requirement.

ID	NAME	WHAT CHANGES
RF-04	Permissions by role	It must include the fourth role, the academic office

3.2 Affected baseline items

Table 3. Affected items of Baseline v2.

ITEM	CHANGE	WHAT HAPPENS
System modules	Added	Module 9 is added, school statistics
App roles	Added	The academic office role is added
Module 1. Users and login	Modified	The new role and its permissions are added
Module 3. Homework	Modified	It must save the date of each result
Module 6. Teacher reports	No change	Module 9 reuses what this one already calculates
Scope	Modified	From 8 modules to 9
Estimated time	Modified	Sprint 11 is added
Budget	Modified	The total cost goes up
Project team	No change	The current team covers it

4. IMPACT ANALYSIS

Table 4. Impact of the change by factor.

FACTOR	IMPACT	ORIGIN	SWOT
People	The current team covers it. No hiring and no layoffs	Internal	Strength
Infrastructure	No change. The current server and database support it	Internal	Strength
Time	A 2-week sprint is added. Delivery moves to April 15, 2027	Internal	Weakness
Budget	It goes up \$27,280 MXN. It is new scope, it was not planned	Internal	Weakness
Client (contract)	It depends on the client signing the contract annex	External	Threat
Business opportunity	The school may ask for more modules and the module helps with other schools	External	Opportunity

People

Programmer 3 already builds the teacher reports and Programmer 1 the screens of the app. Module 9 uses the same skills, so nobody needs to be hired.

Infrastructure

Module 9 makes queries over information the system already saves. It does not create a new volume of data that would need a bigger server.

Time

The original project has 5 months of development plus 1 month of extra time. Using that month to absorb the change was considered, but it was not chosen.

That extra month is a reserve for problems in the development, like a module that is harder than planned or a technical failure. If it is used for extra work asked by the client, the project has no margin left and any later delay becomes a missed delivery date.

For this reason the change is planned as an extra sprint and the delivery moves two weeks. The extra month stays for its original purpose.

Budget

Table 5. Cost of the change.

CONCEPT	TIME	AMOUNT
Programmer 3 (backend, queries and reports)	1 week	\$7,500
Programmer 1 (mobile, screens of the module)	1 week	\$10,500
QA engineer	3 days	\$3,300
Salaries subtotal		\$21,300
Other proportional expenses		\$3,500
Company profit (10%)		\$2,480
Cost of the change		\$27,280 MXN

A 10% profit is used instead of the 25% of the original project. The change is built with the team and the infrastructure already assigned, so it does not create start-up costs for the company.

Table 6. Project comparison.

CONCEPT	BASELINE v2	WITH THE CHANGE
Modules	8	9
Duration	6 months	6 and a half months
Delivery date	March 31, 2027	April 15, 2027
Total cost	\$1,233,125	\$1,260,405 MXN

The increase is 2.2% over the original price.

Client (contract)

The signed scope has 8 modules. The change needs an annex with module 9, the new price and the new date. The development cannot start until the annex is signed.

5. SWOT MATRIX

Table 7. SWOT matrix of the change.

STRENGTHS (internal, positive)	WEAKNESSES (internal, negative)
• The team already has the needed skills • Module 9 reuses the calculations of module 6 • The current infrastructure supports it	• The delivery date moves two weeks • The cost of the project goes up \$27,280 • Module 3 does not save the date and must be modified
OPPORTUNITIES (external, positive)	THREATS (external, negative)
• The school may buy more modules later • The module can be used to offer the app to other schools	• The development cannot start until the client signs the annex • The client may not accept the cost or the new date • The report over time needs months of use to be useful

6. RISK ASSESSMENT

Three levels are used to measure the risk of each part of the change. The level depends on who can solve the risk and how much it affects the project.

Table 8. Risk levels.

LEVEL	WHAT IT MEANS
High	The change cannot be done if this is not solved. It depends on a decision of the client or on something outside the control of the team
Medium	It affects the project plan or the work that was already done, but the team can solve it on its own. It does not stop the change
Low	It is solved with the normal work of the team. It does not affect the plan, the cost or what is already built

Table 9. Risk assessment of the change.

CHANGE	RISK	WHY
Module 9 and academic office role	Low	It is built with the averages module 6 already calculates, and module 1 already handles roles and permissions
Change in module 3, save the date of the result	Medium	It touches a module that was already finished and tested. The tests of that module must be done again
Rescheduling of the project plan	Medium	A 2-week sprint is added without using the extra month, which is kept for problems
Report of learning over time	Medium	It works from the delivery, but it does not give useful information until there are several months of use
Extra cost of \$27,280 MXN	High	It is money the client had not planned. If the client does not approve it, the change is not done
Signature of the contract annex	High	The development cannot start without the signature. It depends on the client and it stops the work